



KTU NOTES

The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

Website: www.ktunotes.in

MODEL QUESTIONS

MODULE I

1. Design zener voltage regulator
2. Clamping circuit any types
3. Clipping circuits any type
4. Explain the procedure to draw dc load line and how to fix Q point
5. What are the factors affecting stability of Q point
6. Any one type biasing method
7. Bias compensation techniques using diode and thermistor
8. Draw and explain the ac equivalent circuit of CE amplifier. Role of capacitors used
9. Derive input impedance output impedance and current voltage gain using h-parameter model/hybrid model
10. Problems related with biasing, dc load line

MODULE II

1. Explain the construction and working of JFET N- channel/ P- channel
2. Explain the construction and working of E-MOSFET N- channel/P- channel
3. Draw and explain the drain and transfer characteristics of FET
4. Explain biasing for JFET/EMOSFET
5. With a neat sketch explain CD and CS amplifier.
6. Explain how FET act as resistor and switch
7. Draw small signal/low frequency model of FET and derive the parameters
8. Miller's theorem with miller capacitance effect
9. Problems related with biasing
10. Frequency Response characteristics of BJT and FET amplifier

MODULE III

1. Multistage amplifiers: Different types of coupling -
2. Compare Direct, RC, & transformer coupled amplifiers
3. Explain the operation of Direct, RC, & transformer coupled amplifiers , advantages and disadvantages
4. Explain the working of Class A, Class B, Class AB and Class C power amplifiers,
5. Derive the Conversion efficiency of Class A/class B
6. Notes on distortion in power amplifiers.
7. Compare Positive and negative feedback,
8. Notes on Effect of negative feedback on gain, input impedance, output impedance, band width and distortion,
9. Draw the Basic feedback topologies
10. A transformer coupled class A power amplifier draws a current of 200mA from a collector supply of 10V, when no signal is applied to it. Determine i) Maximum output power ii) Maximum collector efficiency iii) Power rating of the transistor

MODULE IV

1. Define Barkhausen's criterion,
2. Explain with a neat sketch working of BJT based RC phase shift / Wien bridge oscillators, / working of BJT based Hartley/ Colpitts
3. Derivation of frequency of oscillation for the above mentioned oscillators
4. Neat sketch explains the working of Crystal oscillators.
5. Design any above said oscillators. The parameter values will be given. You must know the expressions
6. Draw Basic block diagram of opamp
7. Draw and explain Ideal op-amp transfer characteristics
8. Explain the op-amp parameters, gain, CMRR, slew rate
9. Explain and draw the Equivalent circuit of ideal and practical op-amp,
10. Compare op-amp configurations- Open loop and closed loop configurations,
11. Write any 6 properties of ideal and practical opamp.
12. Draw and explain the open loop configurations of op amp
13. Compare Inverting and non-inverting closed loop amplifiers and derive expression for voltage gain

14. Explain with a neat sketch and waveform, Voltage follower, Summing amplifier, Subtractor,

15. Design an inverting /non inverting closed loop amplifier. Given gain is 6 or any value

16. Explain the modes of operation of op amp

MODULE V

1. Explain the operation of Differential amplifier with single op-amp,

2. Explain with a neat sketch Instrumentation amplifier-derivation of gain

3. Explain Integrators/ Differentiators/ practical Integrators/ practical Differentiators and derive the expression for gain

4. Explain the working of Comparators/ Zero crossing detector,

5. With neat sketch explain non inverting/inverting Schmitt trigger and derive the expression for gain

6. Implement the equation using two op-amps $V_0 = -5V_1 + 2V_2 - 10V_3$

7. Design a Schmitt trigger circuit with $LTP = -5V$ and $UTP = +5V$. Explain its operation

8. Explain with a neat sketch square/ triangular /ramp generator using an op amp. Derive the frequency of oscillation

9. Explain the effect of slew rate on waveform generation.

10. Explain the block diagram and operation of PLL IC 565.

11. Explain the terms capture, lock, free running in PLL

12. Explain how a Monostable multivibrator can be implemented with 555 Timer IC with relevant waveforms and functional diagram.

13. Explain how an Astable multivibrator can be implemented with a 555 Timer IC with relevant waveforms and functional diagram.

14. Problems related to duty cycle and design of Astable/Monostable

15. Explain the operation of Adjustable voltage regulators IC 723 with a functional diagram

16. Explain the operation of Fixed voltage regulators, 78XX and 79XX series

17. In an astable multivibrator using 555, $R_B = 750 \Omega$. Determine the values of R_A and C to generate a 1.0 MHz clock that has a duty cycle of 25%

18. Design a Wein bridge oscillator to generate a sinusoidal waveform of 2 kHz.

19. Design a monostable multivibrator which produces an output pulse width of 1 ms using 555.

Red colour-problems

Blue-if possible study

Pink colour-only robotics students have to study

Ktunotes.in